

Lem Precompacidad Imp Acotacion Uniforme Compactos

Lema 1. Sea $\Omega \subset \mathbb{C}$ un dominio y $\mathcal{F} \subset \mathcal{C}(\Omega)$ precompacto, entonces $\mathcal{F}$ está uniformemente acotado sobre compactos.

Demostración: Supongamos por contradicción que

$$\exists K \subseteq \Omega \text{ compacto} : \forall n \in \mathbb{N} : \exists f_n \in \mathcal{F}, z_n \in K : |f_n(z_n)| > n.$$

Como K es compacto, $\exists z \in K, (z_{n_k})_{k \in \mathbb{N}} \subset (z_n)_{n \in \mathbb{N}}$ tal que $z_{n_k} \xrightarrow[k \rightarrow \infty]{} z$. Por el **lem-carac-precompacidad** como $\mathcal{F}$ es precompacto, $\exists (f_{n_{k_j}})_{j \in \mathbb{N}} \subset (f_{n_k})_{k \in \mathbb{N}} : \exists f \in \mathcal{C}(\Omega) : f_{n_{k_j}} \xrightarrow[j \rightarrow \infty]{\Omega} f$. Entonces, tenemos que $f_{n_{k_j}}(z) \xrightarrow[j \rightarrow \infty]{} f(z)$, pero

$$\left| f_{n_{k_j}}(z) \right| \geq \left| f_{n_{k_j}}(z_{n_{k_j}}) \right| - \left| f_{n_{k_j}}(z) - f_{n_{k_j}}(z_{n_{k_j}}) \right| > n_{k_j} - \left| f_{n_{k_j}}(z) - f_{n_{k_j}}(z_{n_{k_j}}) \right| \xrightarrow[j \rightarrow \infty]{} \infty,$$

lo que es una contradicción. ■

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